

1. What is CSS?

- CSS stands for **Cascading Style Sheets**.
- It is used to style and design HTML web pages.
- It controls the look and feel of websites (colors, fonts, spacing, layouts etc.).

Example:

```
h1 {  
  color: blue;  
  font-size: 28px;  
}
```

This makes all `<h1>` headings blue and size 28px.

Sample Heading

Small live example — h1 styled with the code above.

★ Benefits of CSS:

- ✓ Makes websites attractive and professional.
- ✓ Saves time — write once, use multiple times.
- ✓ Easy to maintain and update.
- ✓ Improves user experience.

2. Types of CSS

★ Inline CSS

- Used inside HTML tags using **style** attribute.
- Affects only that particular element.

Example:

```
<h1 style="color:red;  
font-size:20px;">  
  Hello World  
</h1>
```

This will make this heading red and size 20px.

★ Internal CSS

- Used inside `<style>` tag in the `<head>` section.
- Affects all elements in the page.

Example:

```
<head>  
<style>  
  h1 {  
    color: blue;  
    font-size: 24px;  
  }  
</style>  
</head>
```

This style will apply to all `<h1>` in this page.

★ External CSS

- Used in a separate .css file.
- Linked to HTML using `<link>` tag.
- Best for large websites.

Example:

```
<head>  
<link rel="stylesheet"  
      href="style.css">  
</head>
```

All styles written in style.css file will be applied to this page.

Comparison Table:

Type	Where it is Written	Scope	Best For
Inline CSS	Inside HTML tag (style attribute)	Single element	Small changes
Internal CSS	Inside <style> tag in <head>	Whole page	Medium websites
External CSS	Separate .css file	Multiple pages	Large websites

3. CSS Syntax

Each CSS rule contains a **selector** and a **declaration block**.

```
selector {  
  property: value;  
  property: value;  
}
```

Example:

```
p {  
  color: green;      /* All <p> green */  
  font-size: 18px;  /* Font-size 18px */  
}
```

This paragraph is styled with the example above.

Small live example — the <p> rule applied above.

Syntax Components

- **Selector:** Targets the HTML element.
- **Property:** The style you want to change.
- **Value:** The value of the property.
- **Declaration:** A property and its value.

4. CSS Selectors

Selectors are used to select the HTML elements you want to style.

Example in HTML:

```
<h1 class="title">Hello</h1>  
<p id="intro">Welcome to CSS Notes</p>  
<p>This is a paragraph.</p>
```

Selector	Syntax	Description	Example	Selects
1. Element Selector	element { }	Selects all elements of that type.	p { color: blue; }	All <p> elements
2. Class Selector	.class { }	Selects all elements with a specific class.	.title { color: red; }	All elements with class="title"
3. ID Selector	#id { }	Selects the element with a specific id.	#intro { font-size: 18px; }	Element with id="intro"
4. Universal Selector	* { }	Selects all elements on the page.	* { margin: 0; }	Every element on the page
5. Group Selector	h1, p { }	Selects multiple elements together.	h1, p { color: purple; }	All <h1> and all <p> elements

★ **Note:**

✓ ID should be unique in a page.

✓ Class can be used multiple times.

✓ You can combine selectors for advanced styling.

Example (Grouping):

```
h1, h2, h3 {  
  color: darkblue;  
}
```

Applies same style to h1, h2 and h3.

Small live example — selector types:

p selector → blue text

.title class → red text

#intro id → bold blue

5. CSS Colors

Colors are used to set the **text color**, **background color**, **border color**, etc.

★ Ways to Specify Colors:

- 1. Color Name — red, blue, green, etc.
- 2. HEX Value — #RRGGBB
- 3. RGB Value — rgb(red, green, blue)
- 4. HSL Value — hsl(hue, saturation%, lightness%)

Small live example — same colors rendered:

color: red;

color: #1E90FF;

rgb(0,128,0);

hsl(120,60%,50%);

Examples:

```
color: red;           /* Color Name */
color: #1E90FF;       /* HEX Value */
color: rgb(0, 128, 0); /* RGB */
color: hsl(120, 60%, 50%); /* HSL */
```

6. CSS Backgrounds

Background properties are used to set the background **color**, **image**, **position**, **repeat**, etc.

Example:

```
body {
  background-color: #f0f8ff;
  background-image: url('bg.jpg');
  background-repeat: no-repeat;
  background-position: center;
  background-size: cover;
}
```

Example Output:



← Background image is set, centered and covers the whole area.

Common Background Properties:

Property	Description
background-color	Sets the background color
background-image	Sets the background image
background-repeat	Repeat the background image
background-position	Sets the starting position of the background image
background-size	Sets the size of the background image
background-attachment	Sets whether background scrolls with the page or not

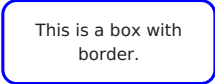
7. CSS Borders

Borders are used to add a **border** around elements.

Example:

```
div {
  border: 2px solid blue;
  border-radius: 10px;
}
```

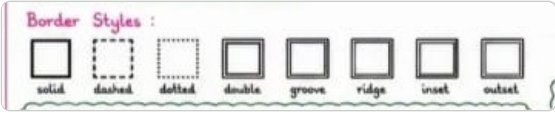
Example Output:



Common Border Properties:

Property	Description
border	Shorthand for width, style and color
border-width	Sets the thickness of the border
border-style	Sets the style (solid, dashed, dotted, etc.)
border-color	Sets the color of the border
border-radius	Rounds the corners of the border

Border Styles:



★ Note:

✓ Color names are not case-sensitive (Red = red = RED).

✓ HEX values must start with #.

✓ Use background-size: cover; for full screen background images.

✓ border-radius: 50% makes the element circular (if width = height).

Example (Circle):

```
div {
  width: 100px;
  height: 100px;
  background: orange;
  border-radius: 50%;
}
```



Output

8. CSS Text & Fonts

Text properties are used to style the text content.

Example:

```
p {
  color: darkblue;
  font-family: 'Poppins', sans-serif;
  font-size: 18px;
  text-align: center;
  text-decoration: underline;
  text-transform: capitalize;
  line-height: 1.6;
}
```

★ **Note:**
Fonts can be from system fonts or web fonts (Google Fonts).

[Sample Paragraph Text Styled Live](#)

Small live example — the rule above applied to real text.

Common Text Properties:

Property	Description	Example
color	Sets the text color	color: red;
font-family	Sets the font of text	font-family: Arial, sans-serif;
font-size	Sets the size of text	font-size: 20px;
font-weight	Sets thickness of text	font-weight: bold;
font-style	Sets style of text	font-style: italic;
text-align	Aligns text (left, center, right, justify)	text-align: center;
text-decoration	Adds decoration (none, underline, overline, line-through)	text-decoration: underline;
text-transform	Changes text case (uppercase, lowercase, capitalize)	text-transform: uppercase;
line-height	Sets space between lines	line-height: 1.8;
letter-spacing	Sets space between letters	letter-spacing: 2px;
word-spacing	Sets space between words	word-spacing: 5px;

9. CSS Lists & Tables

★ A. List Styles

Used to remove or change the default style of lists.

Example:

```
ul {
  list-style-type: square;
  list-style-position: inside;
}
```

Common list-style-type values:

Value	Example
disc	• Item
circle	◦ Item
square	■ Item
none	Item (no bullet)
decimal	1. Item
upper-roman	I. Item
lower-alpha	a. Item

- First item
- Second item

Small live example — square list rendered above.

★ **Quick Tip:**
Good text styling improves readability and makes your website look professional.

★ B. Table Styles

Used to style HTML tables.

Example:

```
table {
  width: 100%;
  border-collapse: collapse;
}
th, td {
  border: 1px solid #4CAF50;
  padding: 10px;
  text-align: center;
}
```

Example Output:

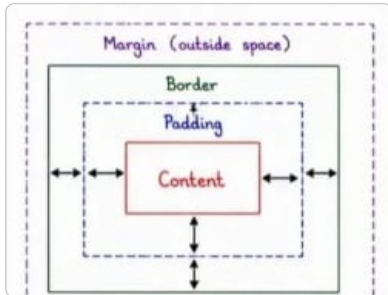
Name	Course	City
Sagar	BCA	Jaipur
Rahul	MCA	Delhi
Neha	B.Sc IT	Pune

★ **Note:**
✓ Use `border-collapse: collapse;` to remove double borders in tables.
✓ `padding` adds space inside table cells.

10. CSS Box Model

Every HTML element is a rectangular box. It consists of:

- 1. Content
- 2. Padding
- 3. Border
- 4. Margin



Example:

```
div {
  width: 300px;
  padding: 20px;
  border: 2px solid blue;
  margin: 15px;
}
```

★ Note:

Margin is outside the border. Padding is inside the border around the content.

11. CSS Borders

Borders are used to define the boundary of an element.

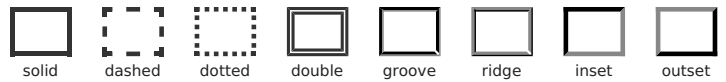
Example:

```
p {
  border-width: 2px;
  border-style: solid;
  border-color: red;
}
```

sample text

Small live example — border-width/style/color applied.

Border Styles:



Border Shorthand:

```
border: 2px dashed green;
```

(width) (style) (color)

★ Note:

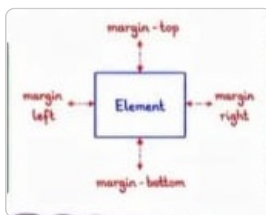
If border-width is not given, it is 3px by default.

12. CSS Margins & Padding

★ Margins (outside space)

Margins create space outside the border.

```
div {
  margin-top: 20px;
  margin-right: 15px;
  margin-bottom: 20px;
  margin-left: 15px;
}
```



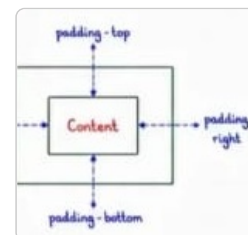
★ Note:

- ✓ You can also use shorthand.
- ✓ margin: 20px; (all four sides)
- ✓ padding: 10px 20px; (top & bottom 10px, left & right 20px)

★ Padding (inside space)

Padding creates space inside the border between content and border.

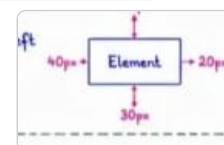
```
div {
  padding-top: 15px;
  padding-right: 10px;
  padding-bottom: 15px;
  padding-left: 10px;
}
```



Shorthand Order (4 values):

margin / padding : top right bottom left

Example: margin: 10px 20px 30px 40px;



13. CSS Display Property

The **display** property specifies how an element is displayed on a webpage.

Common Values:

Value	Description	Example
block	Takes full width. Starts on a new line.	div { display: block; }
inline	Takes only required width. Does not start on new line.	span { display: inline; }
inline-block	Behaves like inline but we can set width & height.	div { display: inline-block; }
none	Hides the element. Not visible on the page.	p { display: none; }
flex	Used for flexible box layout.	div { display: flex; }
grid	Used for grid layout.	div { display: grid; }

★ **Note:**
By default,
div, h1 to h6, p, form, header, footer → display: block
span, a, img, label → display: inline

Example:

```
.box {
  display: inline-block;
  width: 100px;
  height: 50px;
  background-color: lightblue;
  margin: 5px;
  text-align: center;
  line-height: 50px;
}
```

Visual Representation (live):

1. block

Block Element

Block Element

2. inline

Inline

Inline

Inline

3. inline-block

Box

Box

Box

4. none

(Element is hidden)

Output:

Box

Box

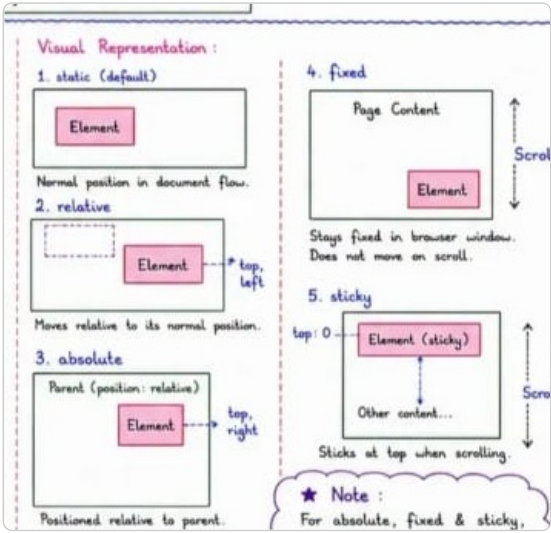
Box

14. CSS Position Property

The **position** property specifies the type of positioning method used for an element.

Value	Description	Example
static	Default position. Follows normal flow.	div { position: static; }
relative	Positioned relative to its normal position.	div { position: relative; top:10px; left:20px; }
absolute	Positioned relative to nearest positioned ancestor. Removed from normal flow.	div { position: absolute; top:0; right:10px; }
fixed	Fixed position relative to browser window. Does not move on scroll.	div { position: fixed; bottom:20px; right:20px; }
sticky	Acts like relative until a certain scroll position, then sticks like fixed.	div { position: sticky; top:0; }

Visual Representation:



Small live example — relative vs absolute:

relative (top:10px; left:20px)

position: relative — shifted from its normal spot.

absolute (top:0; right:10px)

position: absolute — placed relative to the frame.

15. CSS Flexbox (Basics)

Flexbox is a one-dimensional layout method for arranging items in **rows** or **columns**.

Flex Container:

Item 1

Item 2

Item 3

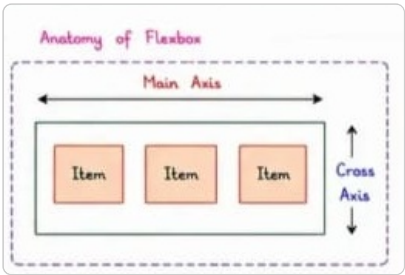
Flex Items

To use flexbox:

```
.container {  
  display: flex;  
}
```

All direct children become flex items.

Anatomy of Flexbox:



Flex Container Properties (Most Important):

Property	Description	Values	Example
flex-direction	Defines the direction of items.	row row-reverse column column-reverse	flex-direction: row;
justify-content	Aligns items along main axis.	flex-start flex-end center space-between space-around space-evenly	justify-content: center;
align-items	Aligns items along cross axis.	stretch flex-start flex-end center baseline	align-items: center;
flex-wrap	Allows items to wrap to next line.	nowrap wrap wrap-reverse	flex-wrap: wrap;
align-content	Aligns multiple lines (when wrap).	stretch flex-start flex-end center space-between space-around space-evenly	align-content: space-between;

Visual Examples — live flexbox demos:

1. flex-direction: row; (default)

1

2

3

2. flex-direction: row-reverse;

3

2

1

3. flex-direction: column;

1

2

3

4. justify-content: space-between;

1

2

3

5. justify-content: center;

1

2

3

6. align-items: center;

1

2

3

7. flex-wrap: wrap;

1

2

3

4

5

Flex Item Properties:

Property	Description	Example
flex-grow	How much item will grow.	flex-grow: 1;
flex-shrink	How much item will shrink.	flex-shrink: 1;
flex-basis	Initial size of item.	flex-basis: 200px;
flex	Shorthand for grow, shrink, basis.	flex: 1 1 200px;
align-self	Align single item (override align-items).	align-self: flex-end;

Example Code:

```
.container {  
  display: flex;  
  justify-content: space-around;  
  align-items: center;  
  height: 150px;  
  border: 2px solid #4CAF50;  
}  
  
.item {  
  background: skyblue;  
  padding: 15px 25px;  
}
```

Output (live):

Item 1

Item 2

Item 3

★ Note:

Flexbox is perfect for building menus, cards, navigation bars, centering elements and modern layouts.

That's it for CSS Basics! Practice more & build amazing websites. Happy Coding!

Remember: "Code + Practice = Skill + Confidence"